

WORKSHOP ON EDA (EXPLORATORY DATA ANALYSIS)

AGENDA:

- 1- What is EDA & Deep understand about EDA
- 2- What is Raw Data
- 3- Missing Value treatment
- 4- What is Clean Data
- 5- Create Clean dataset for predictive model
- 6- Variable Identification
- 7- Univariate Analysis
- 8- Bivariate Analysis
- 9- Outlier Detection
- 10- Variable Creation

PRACTICLE:

- 11- Select Dataset
- 12- We will perform Data preparation
- 13- We will do Data cleaning using PANDAS, NUMPY
- 14- Perform Exploratory Data Analysis using MATPLOTLIB, SEABORN
- 15- Summarization & Documentation

What is EDA →

- EXPLORATORY DATA ANALYSIS (EDA) it is a data exploration technique to understand the various aspect of the data.
- Analyse the data various way – Excel, Python, R, BI tool – Tableau
- Excel data visualization is limited & for bigdata you can't visualize the data in excel
- That's why we need to go for one programming language
- Today we build EDA project end-to-end with practical
- Tableau you can visualize when the data is cleaned
- BI tools you can't visualize RAW DATA
- Let's discuss What is Raw data



RAW DATA →

- The data that has not been processed before.
- Sometime it also called as source data.
- Data directly comes from customer, weblog, html, xml etc.
- Data that is collected directly from the source and hasn't been processed, organized, cleaned or visually presented is considered raw data

Name	Domain	Age	Location	Salary	Exp
Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
Teddy^	Testing	45' yr	Bangalore	10%%000	<3
Uma#r	Dataanalyst^^#			1\$5%000	4> yrs
Jane	Ana^^lytics		Hyderbad	2000^0	
Uttam*	Statistics	67-yr		30000-	5+ year
Kim	NLP	55yr	Delhi	6000^\$0	10+

MISSING VALUE TREATMENT →

- Can't visualize the data if the data has many missing values.
- Data categorized into 2 type – Numerical Data & Categorical Data
- If numerical Data is missing – then we will implement MEAN, MODE, MEDIAN strategy
- If categorical data is missing – then we will implement Mode Strategy

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Numerical Data		Categorical Data	
Missing Data	Fill Missing Data	Missing Data	Fill Missing Data
10	10	Summer	Summer
20	20	Winter	Winter
	20		Winter
30	30	Rainy	Rainy
		Winter	Winter

CLEAN DATA →

- Clean data technique also called as DATA CLEANSING which is very important in DATA ANALYST & DATA SCIENTIST journey.
- Data cleansing or data cleaning is the process of detecting and correcting corrupt or inaccurate records from a record set, table, or database and refers to identifying incomplete, incorrect, inaccurate or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.

Name	Domain	Age	Location	Salary	Exp
Mike	Datascience	34	Mumbai	5000	2
Teddy	Testing	45	Bangalore	10000	3
Umar	Dataanalyst	50	Bangalore	15000	4

VARIABLE IDENTIFICATION →

- Variable identification in the dataset is very very important to choose right machine learning algorithm. Variable or attribute or feature are split into 2 types →
 - Independent Variable | Non target variable | Non predicted variable
 - Dependent Variable | target variable | Predicted variable

I.V	I.V	I.V	I.V	I.V	I.V	I.V	I.V	I.V	D.V
NAME	SFT	OFFICE	SHOPING	SCHOOLS	METROS	PLAYGROUP	SALARY	HOME	Purchase
XYZ	100							Z	Y

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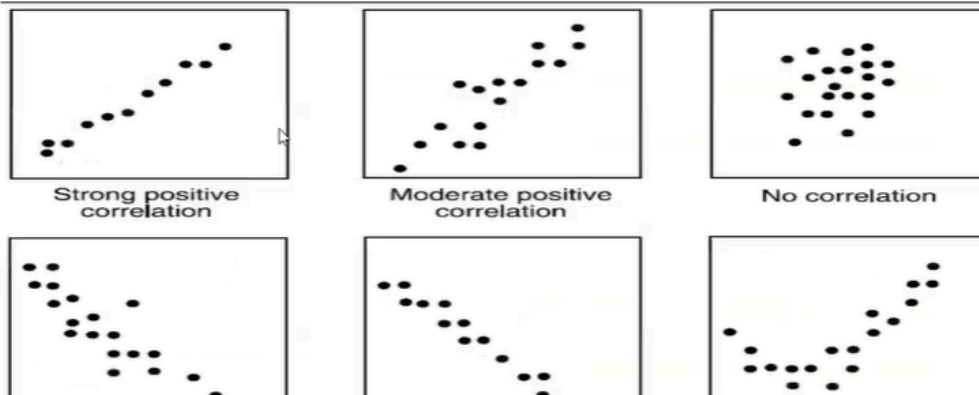
VARIABLE IDENTIFICATION →

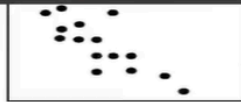
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NAME	SFT	OFFICE	SHOPING	SCHOOLS	METROS	PLAYGROUP	SALARY	HOME	Purchase
XYZ	100							Z	Y
BZC	200	IY	N	N		Y	N	Y	N
ASDF	300		Y	Y	Y		200		Y

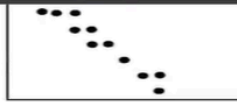
UNIVARIATE & BIVARIATE ANALYSIS →

- Visualize the graph using one variable is called Univariate Analysis
- Visualize the graph using 2 variable is Bivariate Analysis
- Visualize the graph using more than 2 variable or many variables is Multivariate analysis
- Relation Between 2 variable – **CORELATION**
- Below is the pattern of corelation. Corelation is ranging from -1 to 1
- 0 to 1 → Positive corelation
- -1 to 0 → Negative Corelation
- 0 → No Corelation

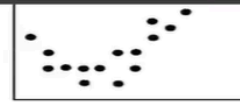




Moderate negative correlation



Strong negative correlation



Curvilinear relationship

3

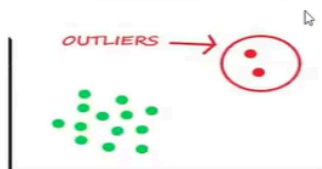
OUTLIER DETECTION →

- Outlier detection also called as Anomaly Detection. This is very important in model building
- Outlier will impact the many classifications algorithm. Eg: Logistic Regression & KNN algorithm
- In statistics, an outlier is a data point that differs significantly from other observations.

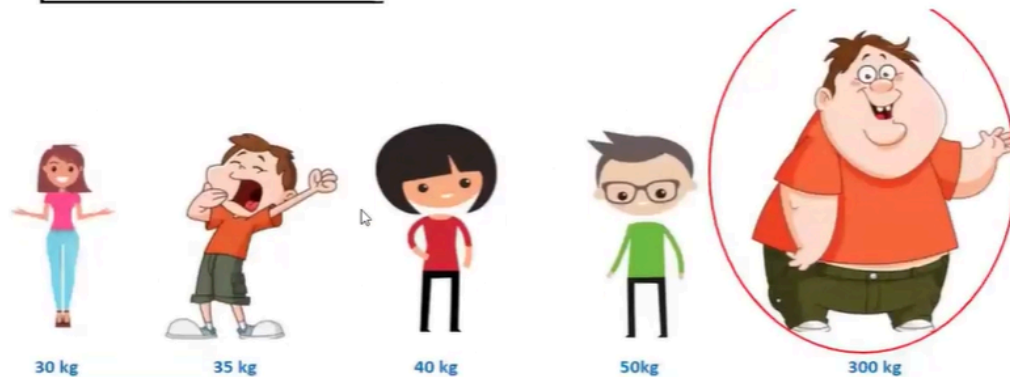
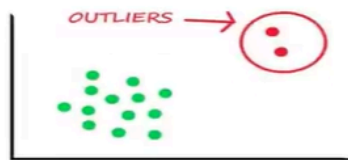
OUTLIERS → 

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VARIABLE CREATION →

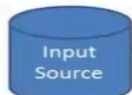
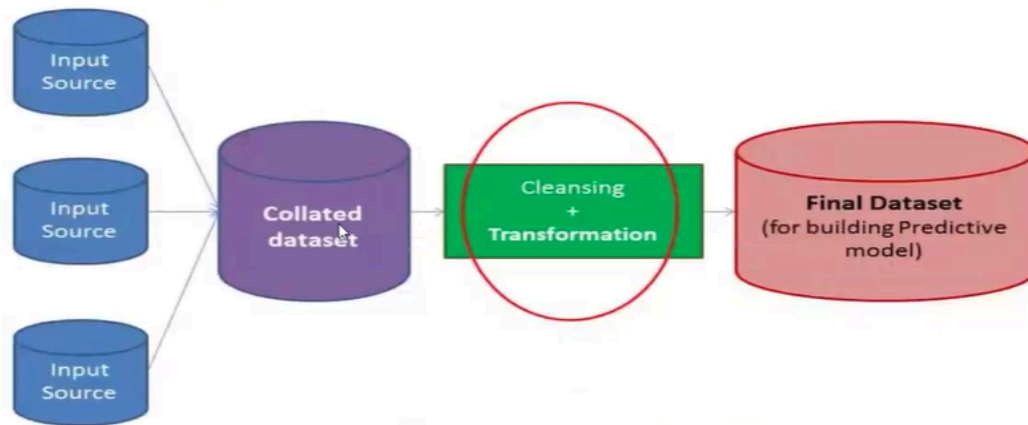
30 kg 35 kg 40 kg 50kg 300 kg

VARIABLE CREATION →

- While we build Machine learning model machine does not understand Sunny, Winter, Rainy.
- Let's impute categorical data to Numerical data before we training the data.
- Variable Creation also we called Encoding Technique
- Dummy variable is one of encoding Technique

DUMMY VARIABLE			
SEASON	SUNNY	RAINY	WINTER
SUNNY	1	0	0
RAINY	0	1	0
WINTER	0	0	1
SUNNY	1	0	0
WINTER	0	0	1
RAINY	0	1	0
RAINY	0	1	0

Process to create dataset for predictive models



WHAT IS THE NEXT STEP →

- Next step is Machine Learning model building
- Then test the model accuracy
- Pass the future data to the model & model generate future prediction
- 1st level of test case to compare future prediction with live data
- Continue 3 level of test
- If all test cases are pass then we need to go for deployment
- After deployment we need to retrain the model with new data
- Final step is to Automize the ML model

SUMMARIZATION →

In this workshop we understand

- What is raw data
- How to clean data using python package - Pandas, NumPy
- How to visualize the data using package - matplotlib, seaborn
- How to convert one data type to other data type

EDA (Exploratory data analysis)

- * eda foundation to become machine learning developer
- * if you are not understand eda then -- hard to build ml model
- * if you not build ml model -- then mlops will not undersnad
- * mlops not understand - cicd pipeline you never building

In order to understand the EDA we need to read the below scenario.

[Input Source 1] ---\
[Input Source 2] ----> [Collated dataset] --> [Cleansing + Transformation] --> [Final Dataset]
(for building Predictive model)
[Input Source 3] ---/

Everyday customers are going to Dmart when ever customers purchase items records are created that records store in dmart database Dmart has its local DB

When we go to reliance it has relaince DB similarly when we go to walmart it has walmart DB.
In all this we have the Raw data

RawData : It means the data which has the missing values is called the raw data.

ex : if we go to more supermarket we will purchase items give mobile number they dont have all the details of me

They is a comapny called Accenture it will reach to all the comapnies Dmart,Walmart,relaince so for this data sharing he will give 3 crores for 2 years sharing data
this is called data sharing.

Every day midnight data transfers from all local DB to Accenture master DB.Choosen midnight because DB is in idel stae no one will work at that time and this process is called ETL Extract Transpose Load it will load data to master DB.

Datawarehouse : The data comes from differnt places and store at one place is called datawarehouse

In the collated dataset it will store raw data someone is there to convert raw data to clean data

Inoder to convert raw data to clean data EDA techniques require.

* Once the data is cleaned we will build ML model

Machine Learning born from EDA technique

1. ****variable identification****
independent variable (non target | non predictive) -- x
dependent variable (target | predictive)----- - y
2. ****univariate analysis****
3. ****bivariate analysis****
4. ****outlier treatment****
5. ****missing value treatment****
6. ****imputation technique (transformer)****
7. ****new variable creation****

variable identification : Column names is called variable identification

columnsnames = feature = attributes

Ex: In a family father, mother, son, daughter so i will considered 4 columns as variables

father | employ ----->Dependent variable = TARGET ATTRIBUTE == PREDICTED
VARIABLE(Y)
mom | house wife----->Independent variable = NonTARGET ATTRIBUTE ==
NonPREDICTED VARIABLE(X1)
son | 3rd grade----->Independent variable = NonTARGET ATTRIBUTE == NonPREDICTED
VARIABLE(X2)
daughter - 2nd grade----->Independent variable = NonTARGET ATTRIBUTE ==
NonPREDICTED VARIABLE(X3)

All 3 members depends on father that is why father or employ called dependent variable.

Note : Dependent variable always taken as Y
Independent variable always taken as X1

So the equation is $Y = X1 + X2 + X3$

$y = m1X1 + m2X2 + m3X3$ -----> This is called multiple linear regression algorithm

The variable identification again divided in to two

1. Independent variable ----->X
2. Dependent variable----->Y

If dependent variable continuous then it is called regression concept.

dv is continuous = = regression concept

Q)What is meant by continuous

gold price ----->continuous increasing
petrol price ----->continuous increasing
stock price ----->continuous increasing
ev price ----->continuous increasing
monthly electricity price ----->continuous increasing

This concept is called regression

To build this above projects we need regression algorithm

- 1.Simple linear regression algorithm
- 2.multiple linear regression algorithm
- 3.gradient descend algorithm
- 4.Stochastic gradient descend algorithm
- 5.batch gradient descend algorithm
- 6.lasso regularization | l1 regularization | l1 regression
- 7- ridge regularization | l2 regularization | l2 regression
- 8- polynomial regression algo
- 9- k nearest neighbour regression algo
- 10- support vector regression algo
- 11- decision tree regression algo
- 12- random forest regression
- 13- xgboost (xtreme gradient boosting)
- 14- lgbm (light gradient boosting)
- 15- time series algorithm
- 16- artificial neural network regression

MACHINE LEARNING is divided into 3 parts

REGRESSION -- Dependent variable is continuous | continuous variable

CLASSIFICATION -- Dependent variable is binary | binary variable

CLUSTERING -- no dependent variable if it is discrete | discrete variable

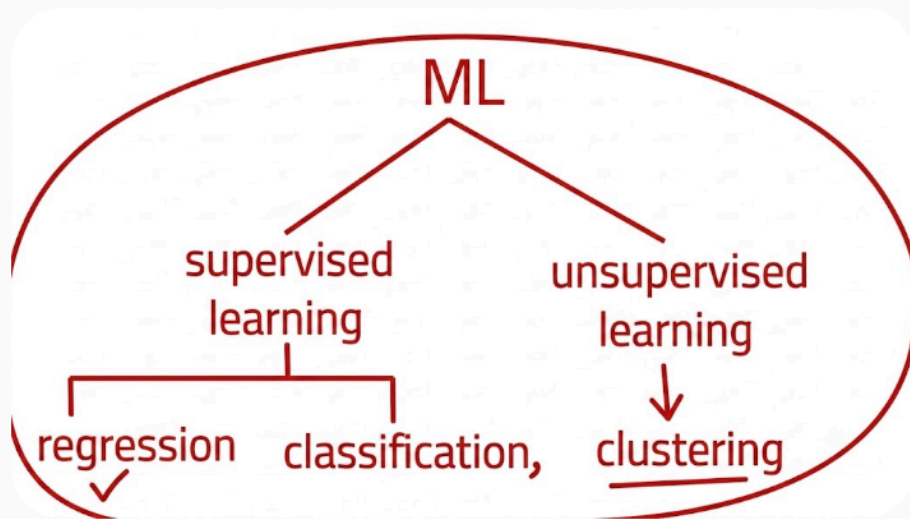
****MACHINE LEARNING****

*** **supervised learning (REGRESSION -- Dependent variable is continuous | continuous variable****

*** **supervised learning (CLASSIFICATION -- Dependent variable is binary | binary variable****

*** **unsupervised learning (CLUSTERING -- no depende variable if is discrest | discrete variable****

In classification DV is binary



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CLASSIFICATION -- Dependent variable is binary | binary variable

Student is pass | fail in exam
 Company is Profit | loss
 Email receive Spam | non Spam
 Restaurent food Tasty | not Tasty
 Stock market is Up | Down
 Purchase | not Purchase
 Male | Female
 True | False
 Yes | No
 Win | Lose

* For this uscase we need to use classification algorithm

Classification Algorithm

1. Logistic REGRESSION
2. Support Vector Machine
3. Knn Classifier
4. naive bayes algorithm(baysian theorm)
5. decission tree
6. Random Forest
7. Xgboost
8. ann classifier

Clustering :-----> Grouping (no dependent variable)

Example : Rating of the food in restaurent

100 students went to restaurent and asked all the students to rate the food.

CLEAN DATA				
NAME	ITEM	PRICE	PHONE	ADDRE
john	food	3000	123	HYD
max	sport	4000	345	BANG

1cr in bank account									
									y
name	sft	scool	college	hospital	shopping	workplace	bhk	price	gender
abc	100	y	n	y	n	n	2	1cr	m
def	200	n	y	y	y	f	3	2cr	f

Here price is dependent variable in order to fix price we need to check all the remaining attributes

1cr in bank account										
x1	x2	x3	x4	x5	x6	x7	x8	y	x9	x10
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	gender	customer look
abc	100	y	n	y	n	n	2	1cr	m	white
def	200	n	y	y	y	f	3	2cr	f	black

For the above data set we need to apply the regression model because the price is continuously increasing. And also price is dependent variable it is not binary so we apply linear regression

For the above data set if we add one more attribute that is purchase then the price becomes independent variable and the purchase becomes dependent variable becomes binary because purchase | not purchase

1cr in bank account											
x1	x2	x3	x4	x5	x6	x7	x8	x11	x9	x10	Y
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	gender	customer look	purchased
abc	100	y	n	y	n	n	2	1cr	m	white	y
def	200	n	y	y	y	f	3	2cr	f	black	n

In both above sets the irrelevant attributes are X9 and X10.

We can remove irrelevant attributes

x1	x2	x3	x4	x5	x6	x7	x8	x9	Y
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	purchased
abc	100	y	n	y	n	n	2	1cr	y
def	200	n	y	y	y	f	3	2cr	n

We will use below set to build machine learning module

x1	x2	x3	x4	x5	x6	x7	x8	x9	Y
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	purchased
abc	100	y	n	y	n	n	2	1cr	y
def	200	n	y	y	y	f	3	2cr	n

Because when we use more attributes then we will get problem of over fitting or multicollinearity problem

1cr in bank account											
x1	x2	x3	x4	x5	x6	x7	x8	x11	x9	x10	Y
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	gender	customer look	purchased
abc	100	y	n	y	n	n	2	1cr	m	white	y
def	200	n	y	y	y	f	3	2cr	f	black	n

Relanvant attribute											
x1	x2	x3	x4	x5	x6	x7	x8	x9	Y		
name	sft	scool	college	hospital	shopping	workplace	bhk	Price	purchased		
abc	100	y	n	y	n	n	2	1cr	y		
def	200	n	y	y	y	f	3	2cr	n		

multicollinearity or overfitting problem here

1- Variable identification

- independent variable (x1, x2, x3 ----- xn)
- dependent variable (y)
- non relevant attribute
- relevant attribute

Note: every time while building ml model we must have to look on relevant attributes or else overfitting problem will happen

2- univariate analysis -- plot the graph using 1 variable

3- outlier treatment --

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4- bivariate analysis